

LINEAR ALGEBRA

A Reference: Definitions, Theorems, and Notation

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A compact reference for a proof-based first course in linear algebra, in the conventions of Friedberg, Insel & Spence. Statements are given without proof. Throughout, F denotes a field (usually $\mathbb{R}$ or $\mathbb{C}$), and all vector spaces are over F unless stated otherwise.

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I NOTATION

$\mathbb{F}^n$	column vectors $(a_1, \dots, a_n)^t$ with entries in $\mathbb{F}$
$M_{m \times n}(\mathbb{F})$	$m \times n$ matrices over $\mathbb{F}$; A_{ij} is the (i, j) entry
$P_n(\mathbb{F}), \mathcal{P}(\mathbb{F})$	polynomials of degree $\leq n$; all polynomials
$e_1, \dots, e_n$	the standard basis of $\mathbb{F}^n$; δ_{ij} is the Kronecker delta
I_V, I_n	identity map on V ; $n \times n$ identity matrix
$\text{span}(S)$	set of all (finite) linear combinations of elements of S ; $\text{span}(\emptyset) = \{0\}$
$\mathcal{L}(V, W), \mathcal{L}(V)$	linear maps $V \rightarrow W$; linear operators on V
$N(T), R(T)$	null space (kernel) and range (image) of T
$\text{nullity}(T), \text{rank}(T)$	$\dim N(T)$ and $\dim R(T)$
$[v]_\beta$	coordinate vector of v relative to an ordered basis β
$[T]_\beta^\gamma, [T]_\beta$	matrix of T relative to β (domain) and γ (codomain); $[T]_\beta = [T]_\beta^\beta$
L_A	left-multiplication map $x \mapsto Ax, \mathbb{F}^n \rightarrow \mathbb{F}^m$
A^t, A^*	transpose; conjugate transpose $(A^*)_{ij} = \overline{A_{ji}}$
$\det(A), \text{tr}(A)$	determinant; trace $\sum_i A_{ii}$
$W_1 \oplus W_2$	direct sum: $W_1 + W_2$ with $W_1 \cap W_2 = \{0\}$
V^*, T^t	dual space $\mathcal{L}(V, \mathbb{F})$; transpose (dual) map $g \mapsto g \circ T$
V/W	quotient space, cosets $v + W$
E_λ, K_λ	eigenspace $N(T - \lambda I)$; generalized eigenspace
$f(t), p(t)$	characteristic polynomial $\det(A - tI)$; minimal polynomial
$\langle x, y \rangle, \ x\ $	inner product; norm $\sqrt{\langle x, x \rangle}$
$S^\perp$	orthogonal complement $\{x : \langle x, y \rangle = 0 \text{ for all } y \in S\}$
T^*	adjoint of T : $\langle Tx, y \rangle = \langle x, T^*y \rangle$

2 VECTOR SPACES

DEFINITION: VECTOR SPACE. A set V with addition $V \times V \rightarrow V$ and scalar multiplication $\mathbb{F} \times V \rightarrow V$ such that for all $x, y, z \in V$ and $a, b \in \mathbb{F}$: (VS1) $x + y = y + x$; (VS2) $(x + y) + z = x + (y + z)$; (VS3) there is $0 \in V$ with $x + 0 = x$; (VS4) each x has y with $x + y = 0$; (VS5) $1x = x$; (VS6) $(ab)x = a(bx)$; (VS7) $a(x + y) = ax + ay$; (VS8) $(a + b)x = ax + bx$.

Standard examples: $\mathbb{F}^n$, $M_{m \times n}(\mathbb{F})$, $P_n(\mathbb{F})$, $P(\mathbb{F})$, and the space $\mathcal{F}(S, \mathbb{F})$ of all functions from a set S to $\mathbb{F}$. Consequences of the axioms: the zero vector and additive inverses are unique, $0x = 0$, $a0 = 0$, and $(-a)x = -(ax)$.

2.1 Subspaces

THEOREM: SUBSPACE TEST. A subset $W \subseteq V$ is a subspace if and only if $0 \in W$, and $x + y \in W$ and $cx \in W$ whenever $x, y \in W$ and $c \in \mathbb{F}$.

Intersections of subspaces are subspaces; unions generally are not. $W_1 \cup W_2$ is a subspace iff $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. The sum $W_1 + W_2 = \{w_1 + w_2\}$ is the smallest subspace containing both, and $V = W_1 \oplus W_2$ iff every $v \in V$ is *uniquely* $w_1 + w_2$.

2.2 Span, independence, bases

DEFINITIONS. S is *linearly dependent* if $a_1u_1 + \dots + a_nu_n = 0$ for some distinct $u_i \in S$ and scalars not all zero; otherwise *linearly independent*. A *basis* is a linearly independent set that spans V . The *dimension* $\dim V$ is the number of vectors in any basis.

THEOREM: REPLACEMENT THEOREM. If $V = \text{span}(G)$ with $|G| = n$ and $L \subseteq V$ is linearly independent with $|L| = m$, then $m \leq n$, and there is $H \subseteq G$ with $|H| = n - m$ such that $L \cup H$ spans V .

Consequences, for $\dim V = n < \infty$:

- every basis has exactly n elements; any spanning set has $\geq n$ vectors and contains a basis;
- any linearly independent set has $\leq n$ vectors and extends to a basis;
- a set of exactly n vectors is a basis iff it spans iff it is linearly independent;
- $\beta = \{u_1, \dots, u_n\}$ is a basis iff each $v \in V$ is uniquely $\sum a_i u_i$;
- a subspace W has $\dim W \leq n$, with equality iff $W = V$.

Dimension of a sum For finite-dimensional subspaces, $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$. Common dimensions: $\dim \mathbb{F}^n = n$, $\dim M_{m \times n}(\mathbb{F}) = mn$, $\dim P_n(\mathbb{F}) = n + 1$; $P(\mathbb{F})$ is infinite-dimensional.

2.3 Quotient spaces

For a subspace W , the cosets $v + W$ form a vector space V/W under $(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$ and $a(v + W) = av + W$; $v_1 + W = v_2 + W$ iff $v_1 - v_2 \in W$. If V is finite-dimensional, $\dim(V/W) = \dim V - \dim W$.

3 LINEAR TRANSFORMATIONS

DEFINITION: LINEAR MAP. $T: V \rightarrow W$ is linear if $T(x + y) = Tx + Ty$ and $T(cx) = cTx$; equivalently $T(cx + y) = cTx + Ty$. Then $T(0) = 0$. Its *null space* is $N(T) = \{x : Tx = 0\}$ and its *range* is $R(T) = \{Tx\}$; both are subspaces.

THEOREM: RANK–NULLITY (DIMENSION THEOREM). If V is finite-dimensional and $T: V \rightarrow W$ is linear, then $\text{nullity}(T) + \text{rank}(T) = \dim V$.

- T is one-to-one iff $N(T) = \{0\}$.
- If $\dim V = \dim W < \infty$, then T is one-to-one $\iff$ onto $\iff \text{rank } T = \dim V$.
- If β is a basis of V , then $R(T) = \text{span}(T(\beta))$.
- A linear map is determined by its values on a basis, and any assignment of values on a basis extends uniquely to a linear map.

3.1 Matrix representations

For ordered bases $\beta = \{v_1, \dots, v_n\}$ of V and $\gamma = \{w_1, \dots, w_m\}$ of W , write $Tv_j = \sum_i a_{ij}w_i$. Then $[T]_\beta^\gamma = (a_{ij}) \in M_{m \times n}(\mathbb{F})$: column j is $[Tv_j]_\gamma$.

THEOREMS: REPRESENTATION.

$$[T(v)]_\gamma = [T]_\beta^\gamma [v]_\beta, \quad [UT]_\alpha^\gamma = [U]_\beta^\gamma [T]_\alpha^\beta, \quad [T + cU]_\beta^\gamma = [T]_\beta^\gamma + c[U]_\beta^\gamma.$$

The map $T \mapsto [T]_\beta^\gamma$ is an isomorphism $\mathcal{L}(V, W) \cong M_{m \times n}(\mathbb{F})$, so $\dim \mathcal{L}(V, W) = mn$.

3.2 Invertibility and isomorphism

T is invertible if there is U with $UT = I_V$ and $TU = I_W$; then T^{-1} is linear and unique. T is invertible iff it is one-to-one and onto, and in that case $[T^{-1}]_\gamma^\beta = ([T]_\beta^\gamma)^{-1}$. Finite-dimensional V and W are isomorphic iff $\dim V = \dim W$; in particular $V \cong \mathbb{F}^n$ via $v \mapsto [v]_\beta$.

3.3 Change of coordinates

THEOREM: CHANGE OF BASIS. Let β, β' be ordered bases of V and $Q = [I_V]_{\beta'}^\beta$ (its columns are the β -coordinates of the vectors of β'). Then Q is invertible, $[v]_\beta = Q[v]_{\beta'}$, and for $T \in \mathcal{L}(V)$,

$$[T]_{\beta'} = Q^{-1}[T]_\beta Q.$$

Matrices A, B are *similar* if $B = Q^{-1}AQ$ for some invertible Q . Similar matrices share rank, determinant, trace, characteristic polynomial, eigenvalues (with multiplicities), and minimal polynomial.

3.4 Dual spaces

$V^* = \mathcal{L}(V, \mathbb{F})$. If $\beta = \{x_1, \dots, x_n\}$ is a basis of V , the *dual basis* $\beta^* = \{f_1, \dots, f_n\}$ is defined by $f_i(x_j) = \delta_{ij}$, and every $f \in V^*$ satisfies $f = \sum_i f(x_i)f_i$. So $\dim V^* = \dim V$ when finite. The transpose $T^t: W^* \rightarrow V^*$, $T^t(g) = gT$, satisfies $[T^t]_{\gamma^*}^{\beta^*} = ([T]_\beta^\gamma)^t$. The map $x \mapsto \hat{x}$, $\hat{x}(f) = f(x)$, is a natural isomorphism $V \cong V^{**}$ for finite-dimensional V .

4 MATRICES AND LINEAR SYSTEMS

4.1 Elementary operations and rank

The three elementary row (column) operations are: swap two rows; multiply a row by a nonzero scalar; add a multiple of one row to another. Each equals left (right) multiplication by an invertible *elementary matrix*.

THEOREM: RANK. $\text{rank}(A) = \text{rank}(L_A)$ equals the dimension of the column space, which equals the dimension of the row space. Multiplying by invertible matrices preserves rank, so elementary operations preserve rank, and $\text{rank}(A) = \text{rank}(A^t)$. Moreover $\text{rank}(AB) \leq \min\{\text{rank } A, \text{rank } B\}$.

Every matrix of rank r can be reduced by elementary operations to $\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$, and every invertible matrix is a product of elementary matrices. To invert A , row reduce $(A \mid I_n)$ to $(I_n \mid A^{-1})$.

4.2 Systems $Ax = b$

- The solution set K_H of $Ax = 0$ is $N(L_A)$, of dimension $n - \text{rank } A$.
- If s is any solution of $Ax = b$, the full solution set is $s + K_H$.
- $Ax = b$ is consistent iff $\text{rank}(A) = \text{rank}(A \mid b)$.
- If A is $n \times n$, then $Ax = b$ has a unique solution for every b iff A is invertible.
- Gaussian elimination: row reduce $(A \mid b)$ to reduced row echelon form; the RREF of a matrix is unique.

The invertible matrix theorem For $A \in M_{n \times n}(\mathbb{F})$ the following are equivalent: A is invertible; $\text{rank } A = n$; $N(L_A) = \{0\}$; L_A is onto; the columns (or rows) of A form a basis of $\mathbb{F}^n$; $Ax = b$ has a unique solution for every b ; A is a product of elementary matrices; A row reduces to I_n ; $\det A \neq 0$; 0 is not an eigenvalue of A .

5 DETERMINANTS

THEOREM: CHARACTERIZATION. $\det: M_{n \times n}(\mathbb{F}) \rightarrow \mathbb{F}$ is the unique function that is linear in each row, is zero whenever two rows are equal (alternating), and satisfies $\det(I_n) = 1$. Explicitly,

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) A_{1\sigma(1)} A_{2\sigma(2)} \cdots A_{n\sigma(n)},$$

and by cofactor expansion along row i , $\det(A) = \sum_j (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij})$, where $\tilde{A}_{ij}$ deletes row i and column j .

- Swapping two rows negates $\det$; scaling a row by k scales $\det$ by k ; adding a multiple of a row leaves $\det$ unchanged.
- $\det(AB) = \det(A) \det(B)$, $\det(A^t) = \det(A)$, and $\det(A^{-1}) = \det(A)^{-1}$.
- A is invertible iff $\det A \neq 0$. The determinant of a triangular matrix is the product of its diagonal entries.
- Cramer's rule: if $\det A \neq 0$, the solution of $Ax = b$ is $x_k = \det(M_k) / \det(A)$, where M_k replaces column k of A by b .
- Over $\mathbb{R}$, $|\det A|$ is the factor by which L_A scales n -dimensional volume; the sign records orientation.

6 EIGENVALUES AND DIAGONALIZATION

DEFINITIONS. $\lambda \in \mathbb{F}$ is an *eigenvalue* of $T \in \mathcal{L}(V)$ if $Tv = \lambda v$ for some $v \neq 0$, an *eigenvector*. The *characteristic polynomial* of $A \in M_{n \times n}(\mathbb{F})$ is $f(t) = \det(A - tI_n)$, of degree n with leading coefficient $(-1)^n$; that of T is that of any $[T]_\beta$. The *eigenspace* is $E_\lambda = N(T - \lambda I)$.

- λ is an eigenvalue iff $f(\lambda) = 0$. The *algebraic multiplicity* of λ is its multiplicity as a root of f ; the *geometric multiplicity* is $\dim E_\lambda$, and $1 \leq \dim E_\lambda \leq$ algebraic multiplicity.
- Eigenvectors for distinct eigenvalues are linearly independent.
- If f splits, $\det A = \prod \lambda_i$ and $\text{tr } A = \sum \lambda_i$ (with multiplicity).

THEOREM: DIAGONALIZABILITY. T is *diagonalizable* if V has a basis of eigenvectors of T , i.e. $[T]_\beta$ is diagonal for some β . This holds iff (1) f splits over $\mathbb{F}$ and (2) for each eigenvalue, the geometric multiplicity equals the algebraic multiplicity; equivalently, $V = E_{\lambda_1} \oplus \dots \oplus E_{\lambda_k}$. In particular, n distinct eigenvalues suffice. Then $A = QDQ^{-1}$, where the columns of Q are eigenvectors and D is diagonal, and $A^m = QD^mQ^{-1}$.

6.1 Invariant subspaces and Cayley–Hamilton

A subspace W is *T-invariant* if $T(W) \subseteq W$; then the characteristic polynomial of $T|_W$ divides that of T . The *T-cyclic subspace* generated by $x \neq 0$ is $W = \text{span}\{x, Tx, T^2x, \dots\}$; if $\dim W = k$, then $\{x, \dots, T^{k-1}x\}$ is a basis and $T^k x = -\sum_{i=0}^{k-1} a_i T^i x$ gives the characteristic polynomial $(-1)^k(a_0 + a_1 t + \dots + a_{k-1} t^{k-1} + t^k)$ of $T|_W$.

THEOREM: CAYLEY–HAMILTON. If f is the characteristic polynomial of T , then $f(T) = T_0$, the zero operator. Likewise $f(A) = 0$ for matrices.

6.2 Minimal polynomial

The *minimal polynomial* p of T is the unique monic polynomial of least positive degree with $p(T) = T_0$.

- $g(T) = T_0$ iff p divides g ; in particular $p \mid f$.
- p and f have the same roots: the eigenvalues of T .
- T is diagonalizable iff p splits over $\mathbb{F}$ with no repeated roots, i.e. $p(t) = (t - \lambda_1) \dots (t - \lambda_k)$ for distinct λ_i .

7 CANONICAL FORMS

DEFINITION: GENERALIZED EIGENVECTORS. $x \neq 0$ is a *generalized eigenvector* for λ if $(T - \lambda I)^p x = 0$ for some $p \geq 1$. The *generalized eigenspace* $K_\lambda = \{x : (T - \lambda I)^p x = 0 \text{ for some } p\}$ is T -invariant and contains E_λ .

THEOREM: JORDAN CANONICAL FORM. If the characteristic polynomial of T splits, with distinct eigenvalues $\lambda_1, \dots, \lambda_k$ of multiplicities $m_1, \dots, m_k$, then $\dim K_{\lambda_i} = m_i$, $K_{\lambda_i} = N((T - \lambda_i I)^{m_i})$, and $V = K_{\lambda_1} \oplus \dots \oplus K_{\lambda_k}$. There is an ordered basis (a union of cycles of generalized eigenvectors) in which

$$[T]_\beta = \begin{pmatrix} J_1 & & \\ & \ddots & \\ & & J_r \end{pmatrix}, \quad J = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix}.$$

This *Jordan form* is unique up to the order of the blocks; two matrices with split characteristic polynomials are similar iff they have the same Jordan form.

- The number of Jordan blocks for λ equals $\dim E_\lambda$.
- The number of blocks for λ of size $\geq j$ equals $\text{rank}((T - \lambda I)^{j-1}) - \text{rank}((T - \lambda I)^j)$ (dot diagrams).
- In the minimal polynomial, the exponent of $(t - \lambda)$ is the size of the largest Jordan block for λ .

Over fields where f need not split, the *rational canonical form* plays the same role, built from companion matrices of powers of irreducible factors of f .

8 INNER PRODUCT SPACES

DEFINITION: INNER PRODUCT. On V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, an inner product $\langle \cdot, \cdot \rangle$ satisfies, for all x, y, z and c : $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$; $\langle cx, y \rangle = c\langle x, y \rangle$; $\overline{\langle x, y \rangle} = \langle y, x \rangle$; $\langle x, x \rangle > 0$ for $x \neq 0$. It is therefore conjugate-linear in the second argument: $\langle x, cy \rangle = \bar{c}\langle x, y \rangle$. The standard inner product on $\mathbb{F}^n$ is $\langle x, y \rangle = \sum_i x_i \bar{y}_i = y^* x$; on $M_{n \times n}(\mathbb{F})$ it is $\langle A, B \rangle = \text{tr}(B^* A)$.

THEOREM: BASIC INEQUALITIES. For $\|x\| = \sqrt{\langle x, x \rangle}$: Cauchy-Schwarz $|\langle x, y \rangle| \leq \|x\| \|y\|$; triangle inequality $\|x + y\| \leq \|x\| + \|y\|$; Pythagoras $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ when $x \perp y$.

8.1 Orthonormal bases and Gram-Schmidt

A set is *orthonormal* if its vectors are pairwise orthogonal unit vectors; orthogonal sets of nonzero vectors are linearly independent. If $\beta = \{v_1, \dots, v_n\}$ is an orthonormal basis, then

$$x = \sum_{i=1}^n \langle x, v_i \rangle v_i, \quad \|x\|^2 = \sum_i |\langle x, v_i \rangle|^2, \quad ([T]_\beta)_{ij} = \langle T v_j, v_i \rangle.$$

THEOREM: GRAM-SCHMIDT. If $\{w_1, \dots, w_n\}$ is linearly independent, define $v_1 = w_1$ and

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} v_j \quad (2 \leq k \leq n).$$

Then $\{v_1, \dots, v_n\}$ is orthogonal with the same span (indeed $\text{span}\{v_1, \dots, v_k\} = \text{span}\{w_1, \dots, w_k\}$ for each k); normalizing gives an orthonormal basis. Every finite-dimensional inner product space has an orthonormal basis.

8.2 Orthogonal complements and projections

- For a finite-dimensional subspace W : $V = W \oplus W^\perp$, and $(W^\perp)^\perp = W$ when V is finite-dimensional.
- Each y is uniquely $u + z$ with $u \in W$, $z \in W^\perp$; if $\{v_i\}$ is an orthonormal basis of W , then $u = \sum \langle y, v_i \rangle v_i$.
- *Best approximation*: u is the unique vector in W closest to y , i.e. $\|y - u\| < \|y - x\|$ for all $x \in W$, $x \neq u$.
- If V is finite-dimensional, $\dim W + \dim W^\perp = \dim V$.

Least squares For $A \in M_{m \times n}(\mathbb{F})$ and $y \in \mathbb{F}^m$, the vectors x_0 minimizing $\|Ax - y\|$ are exactly the solutions of the *normal equations* $A^* A x_0 = A^* y$. If $\text{rank } A = n$, then $A^* A$ is invertible and $x_0 = (A^* A)^{-1} A^* y$.

8.3 The adjoint

THEOREM: RIESZ REPRESENTATION AND ADJOINTS. If V is finite-dimensional and $g \in V^*$, there is a unique $y \in V$ with $g(x) = \langle x, y \rangle$ for all x . Consequently every $T \in \mathcal{L}(V)$ has a unique *adjoint* T^* with $\langle Tx, y \rangle = \langle x, T^*y \rangle$, and for an orthonormal basis β , $[T^*]_\beta = ([T]_\beta)^*$.

Rules: $(T + U)^* = T^* + U^*$, $(cT)^* = \bar{c}T^*$, $(TU)^* = U^*T^*$, $T^{**} = T$, $I^* = I$; $N(T^*) = R(T)^\perp$ and $\text{rank } T^* = \text{rank } T$.

9 NORMAL, SELF-ADJOINT, AND UNITARY OPERATORS

DEFINITIONS. On a finite-dimensional inner product space, T is *normal* if $TT^* = T^*T$; *self-adjoint* (Hermitian) if $T = T^*$; *unitary* ($F = \mathbb{C}$) or *orthogonal* ($F = \mathbb{R}$) if $TT^* = T^*T = I$, i.e. $\|Tx\| = \|x\|$ for all x . A matrix A is unitary iff its columns form an orthonormal basis of $\mathbb{F}^n$, iff $A^{-1} = A^*$.

THEOREM: SCHUR. If the characteristic polynomial of T splits, there is an orthonormal basis β for which $[T]_\beta$ is upper triangular. In matrix form: $A = PUP^*$ with P unitary (orthogonal) and U upper triangular.

THEOREM: SPECTRAL THEOREMS. *Complex:* T is normal $\iff V$ has an orthonormal basis of eigenvectors of T .

Real: T is self-adjoint $\iff V$ has an orthonormal basis of eigenvectors of T .

In either case, with orthogonal projections T_i onto the eigenspaces $W_i = E_{\lambda_i}$,

$$T = \lambda_1 T_1 + \dots + \lambda_k T_k, \quad I = T_1 + \dots + T_k, \quad T_i T_j = \delta_{ij} T_i, \quad V = W_1 \oplus \dots \oplus W_k.$$

- Eigenvalues of a self-adjoint operator are real; those of a unitary operator have absolute value 1.
- If T is normal, $\|Tx\| = \|T^*x\|$, and $Tx = \lambda x$ implies $T^*x = \bar{\lambda}x$; eigenvectors for distinct eigenvalues are orthogonal.
- A real symmetric matrix is orthogonally diagonalizable: $A = QDQ^t$ with Q orthogonal and D real diagonal.

9.1 Positive operators and quadratic forms

T is *positive semidefinite* if $T = T^*$ and $\langle Tx, x \rangle \geq 0$ for all x , and *positive definite* if moreover $\langle Tx, x \rangle > 0$ for $x \neq 0$; equivalently, T is self-adjoint with nonnegative (positive) eigenvalues. Every positive semidefinite T has a unique positive semidefinite square root, and A^*A is always positive semidefinite. A real quadratic form $q(x) = x^t Ax$ with A symmetric is positive definite iff all eigenvalues of A are positive; by Sylvester's law of inertia, the numbers of positive, negative, and zero eigenvalues are invariants under congruence $A \mapsto Q^t A Q$.

10 THE SINGULAR VALUE DECOMPOSITION

THEOREM: SVD. Let $A \in M_{m \times n}(F)$ have rank r . There are unitary (orthogonal) $U \in M_{m \times m}$, $V \in M_{n \times n}$ and $\Sigma \in M_{m \times n}$ with $\Sigma_{ii} = \sigma_i$ for $i \leq r$ and all other entries zero, such that

$$A = U\Sigma V^*, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0.$$

The *singular values* σ_i are the square roots of the nonzero eigenvalues of A^*A ; the columns of V (right singular vectors) are an orthonormal eigenbasis of A^*A , and $u_i = \sigma_i^{-1} A v_i$ for $i \leq r$.

- Geometrically, L_A maps the unit sphere to an ellipsoid with semi-axes $\sigma_i u_i$: rotate (V^*), stretch (Σ), rotate (U).
- $\{v_{r+1}, \dots, v_n\}$ is an orthonormal basis of $N(L_A)$ and $\{u_1, \dots, u_r\}$ one of $R(L_A)$.
- *Pseudoinverse*: $A^\dagger = V \Sigma^\dagger U^*$, where $\Sigma^\dagger$ is $n \times m$ with entries $1/\sigma_i$. Then $x = A^\dagger b$ is the least-squares solution of $Ax = b$ of minimal norm.
- *Polar decomposition*: every square A factors as $A = WP$ with W unitary and $P = (A^* A)^{1/2}$ positive semidefinite.

II QUICK REFERENCE

	nullity T + rank T = $\dim V$
DIMENSION THEOREM	
	column j of $[T]_\beta^\gamma$ is $[T v_j]_\gamma$
MATRIX OF A MAP	
	$[T]_{\beta'} = Q^{-1} [T]_\beta Q$, $Q = [I]_{\beta'}^\beta$
CHANGE OF BASIS	
	multilinear, alternating, $\det I = 1$; $\det(AB) = \det A \det B$
DETERMINANT	
	f splits and geometric = algebraic multiplicity; p has distinct roots
DIAGONALIZABLE	
	$f(T) = 0$; $p \mid f$
CAYLEY–HAMILTON	
	$V = \bigoplus K_\lambda$; #blocks for λ = $\dim E_\lambda$
JORDAN FORM	
	$v_k = w_k - \sum_{j < k} \frac{\langle w_k, v_j \rangle}{\ v_j\ ^2} v_j$
GRAM–SCHMIDT	
	$\text{proj}_W y = \sum \langle y, v_i \rangle v_i$ for orthonormal $\{v_i\}$
PROJECTION	
	$A^* A x = A^* y$
LEAST SQUARES	
	$\langle T x, y \rangle = \langle x, T^* y \rangle$, $[T^*]_\beta = [T]_\beta^*$ (β orthonormal)
ADJOINT	
	normal ($\mathbb{C}$) / self-adjoint ($\mathbb{R}$) $\iff$ orthonormal eigenbasis
SPECTRAL THEOREM	
	$A = U \Sigma V^*$, $\sigma_i = \sqrt{\lambda_i(A^* A)}$
SVD	